

Athira Krishnan R

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PROFESSIONAL EXPERIENCE

Tata Consultancy Services

Bangalore, Trivandrum

AI Lead, TCS RnI Incubation/ Systems Engineer

2019-Present (On Leave)

- Worked on autonomous parking for a custom four-wheeled robot using AprilTags and ROS.
- Developed and analyzed BCI-based use cases using NeuroSky EEG signals to study human behavior patterns.
- Built pipelines for PID extraction from bills, user emotion recognition from images, and EEG signal analysis.
- Designed pose-based wellness-oriented games and performed NLP-based information extraction from customer documents.
- Implemented speaker diarization systems and deployed projects on AWS with database management.
- Played key role in setting up BCI lab for Karyavattom University, Trivandrum.
- Served as a mentor for multiple cohort training sessions.

Asimov Robotics

Kalamassery

Asst. Robotic Software Engineer

2018-2019

- Worked on simulations of Saaya, humanoid robot.
- Ensured human-like interactive eye movement in Saaya.
- Worked on navigation and mapping for Saaya and was part of few deployment.
- Worked on planning and trajectory tracking 7DOF robot surgical arms in simulation and real.

Ernst n Young

Kinfra Park, Trivandrum

Intern

2018

- Tested localization on Parrot drone with opensource SLAM.
- Path planning and trajectory tracking in Parrot drone.
- Localizing Parrot drone indoor with Aruco markers.

EDUCATION

PhD in AI

June 2022 - Present

IIT Hyderabad

CGPA-9.25

- Guided by Prof Sumohana Channappayya & Prof Sri Rama Murty K
- Thesis : Multimodal intelligence towards robot navigation

M. Tech in ECE - Specialization: Robotics and Automation

2016 - 2018

College of Engineering Trivandrum

CGPA-9.43

- Thesis : Path planning and trajectory tracking in quadcopter based delivery systems

B. Tech in ECE

2012 - 2016

Mar Baselios College of Engineering and Technology, Trivandrum

CGPA-9.08

- Thesis : Smart Vehicle Modules

PUBLICATIONS

- **AV-IO: Audio-Visual Indoor-Outdoor Classification Dataset via Multimodal Weak Supervision** 2026
Athira Krishnan R, Sumohana S. Channappayya
National Conference on Communications (IEEE NCC)
- **TEPEN: Towards an Ensemble Model for Pixel-based Embodied Navigation** 2025
Athira Krishnan R, Sumohana S. Channappayya
Pattern Recognition and Machine Intelligence (PREMI)
- **OpenVLM-Nav: Training-Free Zero-Shot Object-Goal Navigation via Vision-Language Guidance** 2025
Athira Krishnan R, Swapnil Bag
In Proceedings of the ICC 2025 Workshop on Robotics and Autonomous Systems.

- **Glimpse of MCQ-based VQA in Road & Traffic Scenarios** 2025
Athira Krishnan R, Sumukha BG, Ambarish Parathasarthi
LLVM-AD Workshop, WACV
- **“Systems And Methods For Extracting, Digitizing, And Using Engineering Drawing Data”** 2023
Robin Tommy, R Athira Krishnan, *et al.*
Indian Patent 202121040056, published March 10, 2023.
- **”Method And System For Assessing And Improving Wellness Of Person Using Body Gestures”.** 2023
Robin Tommy, R Athira Krishnan, *et al.*
Indian Patent 202121039878, published March 3, 2023.
- **Experimental Validation of Findings Using BCI in Autistic Children (EVAL-BCI)** 2020
R. Ravindranathan, R. Tommy, Athira Krishnan R
IEEE TENCON
- **Self-Supervised Learning of Robot Manipulation** 2020
R. Tommy, Athira Krishnan R
International Conference on Advanced Computing and Robotics (ICACR)
- **Path Planning of an Autonomous Quadcopter-Based Delivery System** 2018
Athira Krishnan R, V. R. Jisha, K. Gokulnath
ICETIETR

PROJECTS

- Cars Project - Vision for Robot, DRDO** | *Python, ROS2, Pytorch, ONNX, Tensorrt, Edge* Jun 2022 – Present
- Lead and coordinate the team for all the signed modules from vision part.
 - Designed workflow and developed the ROS2 integration of solutions.
 - Developed solutions towards segmentation, action recognition, indoor-outdoor classification, etc.
 - Real time performance in edge compute was the addressed.
- AI Living Lab - INRIA** | *React, nodejs, pytorch, Git* May 2018 – May 2020
- Developed and designed the frontend and backend for AI Living lab
 - Developed in collaboration with team from Europe
 - Presented the summary of work online at INRIA annual summit in 2022.
- Buildabot - Prayana IEEE RAS** | *ROS, pytorch, arduino, raspberry pi* Dec 2024 -Jan 2025
- Served as mentor to build a robot from scratch.
 - Vision, audio and motion capability.
 - Featured in most local news

TECHNICAL SKILLS

Languages: Python, C/C++, SQL (Postgres), JavaScript, HTML/CSS
Frameworks: React, Node.js, FastAPI
Developer Tools: Git, Docker, Neurosky, Emotiv, Google Cloud Platform, AWS Cloud, Open AI API, VS Code, Visual Studio, ONNX, Tensorrt, Solid works , Pega (Basics), beautifulsoup
Middleware: ROS1, ROS2

VOLUNTARY PROFESSIONAL ROLES

- GSAC Advisor** 2025-2026
IEEE RAS Kerala Chapter
- Treasurer** 2021-2026
IEEE RAS Kerala Chapter
- Jnt. Secretary** 2021-2022
IEEE RAS Kerala Chapter
- Industry Expert** 2021-2022
CET, Saintgits College of Engineering
Design Curriculum for Robotics, AI

AWARDS

- IEEE RAS Graduate Student Award, 2025
- TCS Maithree Bronze Prize, 2022.
- TCS Star of the month Award, 2022.
- TCS Service and Commitment Award, 2022.
- TCS Special Initiative Award, 2021.
- TCS On the spot Award, 2020.
- TCS Special Initiative Award, 2020.
- TCS LIREL Award 2019.